

Sahil Nitin Mujumdar

Data Scientist

[Sahil Mujumdar](#)[Sahil Mujumdar Github](#)

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Professional Summary

AI/Data Science graduate with experience building AI-powered applications using machine learning, computer vision, and LLM-based architectures. Skilled in Python, FastAPI, PyTorch, and TensorFlow with hands-on experience in RAG pipelines, real-time AI systems, and model deployment. Experienced in developing production-ready AI APIs, data pipelines, and analytics dashboards to solve real-world problems.

EDUCATION

AP SHAH INSTITUTE OF TECHNOLOGY , THANE (UNIVERSITY OF MUMBAI)

Bachelor of Engineering (B.E), Data Science

Maharashtra, India

July 2021- May 2025

CGPA: 7.4/10.0**P. O. NAHATA COLLEGE OF SCIENCE COMMERCE AND ARTS(BHUSAWAL)**ISC (XII) – Science - **Percentage : 86.03**

Maharashtra, India

May 2019- May 2021

SKILLS SUMMARY

- **Languages:** Python, SQL, Java, R , HTML, CSS
Big Data: Apache Spark, Hadoop (familiar)
AI/ML Domains: TensorFlow, PyTorch, Keras, Scikit-learn, Computer Vision, NLP, (RAG)
Cloud: Apache Spark (familiar), Hadoop (familiar), AWS, GCP, CI/CD pipelines
Ethical AI & Compliance: Data privacy, governance, and security practices
- **Frameworks:** TensorFlow, PyTorch, Keras, Scikit-learn, FastAPI, Flask, OpenCV
- **AI/LLM Tools:** FAISS, RAG (Retrieval Augmented Generation), Vector Embeddings
- **Database:** MySQL, SSMS, MongoDB
- **Version Control:** Git, Github, Bitbucket
- **Soft Skills:** Mentor teammates, resolved technical challenges, developed data-driven Strategies, communicated technical concepts effectively.

EXPERIENCE

Tobox Ventures | On-site**July 25 – Present****Jr. Data Scientist**

- **Assisting in the development of predictive models and data pipelines using Python, SQL, and ML frameworks to support business insights.**
- **Conducting data preprocessing, visualization, and statistical analysis to identify trends and improve decision-making efficiency.**

National Informatics Centre (NIC). | Hybrid**July 24 – June 25****Data Analyst Intern**

- **Developed an Election Dashboard:** Designed a highly interactive UI/UX using ASP.NET with Chart.js for data visualization and SQL Server for database integration, resulting in a 30% improvement in data accessibility and decision-making efficiency
- **Conducted Security Audits:** Performed comprehensive website security audits, identifying and mitigating vulnerabilities, which enhanced system security by 40% and ensured compliance with industry standards.
- **Built Web Templates and Systems:** Created reusable HTML templates for seamless usability and implemented Redis server functionality, reducing application response time by 25% and improving user experience.

Zscaler. | Remote**July 24 - Sept 24****Networking Virtual Internship**

- Gained hands-on experience with Zscaler's Zero Trust Exchange platform, improving understanding of secure cloud transformations and zero-trust architecture.
- Identified and mitigated vulnerabilities in simulated environments, boosting network resilience and practical troubleshooting capabilities by 20%.
- Designed and optimized policies for secure web gateway, SSL inspection, and cloud firewalls, ensuring secure internet access.

PROJECTS

Blockchain Based Ammunition System

June 24

Pioneered a secure blockchain solution for ammunition logistics, ensuring tamper-proof data integrity. Demonstrated leadership in research and implementation, aligning with defense and supply chain use cases.

- Designed a secure and transparent ledger using Solidity smart contracts to track ammunition and military logistics, ensuring tamper-proof data integrity and minimizing manual errors.
- Implemented blockchain technology to enable real-time tracking of inventory, orders, and deliveries, optimizing the overall efficiency of the supply chain process.
- Utilized cryptographic techniques and decentralized mechanisms to protect sensitive military data, ensuring secure access and traceability for all stakeholders.

Advanced Ammunition Prediction System

April 24

Directed end-to-end ML lifecycle from data collection to visualization. Optimized procurement and resource allocation using predictive models with diverse data sources.

- Designed a predictive analytics system using machine learning models to forecast ammunition demand based on historical data, ensuring timely procurement and efficient resource allocation.
- Incorporated diverse data sources such as usage patterns, environmental factors, and supply chain metrics to enhance the accuracy of predictions and optimize inventory management.
- Built intuitive dashboards to visualize prediction trends and provide actionable insights, supporting data-driven decision-making for military logistics and operations.

MedAgentX (E-Doctor OS) :Agentic AI Platform for Clinical Decision Support

June25

- Designed an agentic AI platform enabling doctors and healthcare researchers to build programmable medical recommendation systems.
- Implemented multi-agent orchestration with advanced RAG pipelines to support contextual clinical decision assistance.
- Integrated governance and safety guardrails with human-in-the-loop approval architecture for responsible medical AI usage.

OmniTutor AI — Multi-LLM Orchestrator with RAG

Jan 26

- Developed a multi-provider LLM orchestration platform using Fast API that dynamically routes prompts across models such as OpenAI, Groq, and Gemini.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using FAISS to enhance responses with indexed knowledge retrieval.
- Built a Streamlit dashboard for querying, provide monitoring, and usage analytics with circuit breakers and failover mechanisms.

PUBLISHCATIONS

Defense Ledger – Blockchain-Powered Ammunition and Supply Chain — Presented at **6th International Conference of Emerging Technology (INCET 2025)**, Jain College of Engineering, Belagavi, India (May 2025). | [LINK](#)

MedAgentX: Agentic AI Platform for Clinical Decision Support (*Under Preparation / Journal Submission*)

CERTIFICATES

- Networking Virtual Internship from Zscaler | [LINK](#)
- Cloud Foundations from AWS | [LINK](#)
- Kubernetes Foundation from Udemy | [LINK](#)
- Data Engineering Professional | [LINK](#)
- Data Engineering Master | [LINK](#)
- Machine Learning Professional | [LINK](#)
- Machine Learning Master | [LINK](#)
- Power BI Udemy | [LINK](#)